

Estimator choice is the dominant forking path: a specification-curve analysis of panel-data estimation

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Abstract

Specification-curve and multiverse analyses are now standard across the quantitative social and behavioural sciences for gauging how sensitive a result is to defensible analytic choices. In longitudinal (panel) data these analyses almost always vary covariates and samples while holding the *estimator* fixed. We argue this is backwards: for panel data the estimator is the choice of identifying assumption — pooled, between, random-effects, fixed-effects and first-difference estimators use systematically different variation and answer different questions. Using a widely used wage panel (545 individuals observed over 1980–1987), we enumerate the full specification curve of three estimands — the union and marriage wage premiums and the return to schooling — across five estimators crossed with covariate, sample, time-effect and inference choices, and decompose the variance of the estimate distribution by analytic dimension. Estimator choice alone accounts for 84.6% of the variance of the union premium and 56.2% of the marriage premium; covariates and sample together account for under 7%. The return to schooling is unidentifiable under every within estimator. On these data, that pattern recommends treating the estimator as a first-class forking path in panel multiverse analyses. All estimates are reproduced in R and Python, and an open, browser-based implementation reconstructs every result.

Introduction

A growing methodological consensus across the quantitative sciences holds that a single reported estimate is rarely “the” answer: it is one path through a branching set of defensible analytic decisions (Gelman and Loken 2013; Leamer 1983). Specification-curve analysis (Simonsohn et al. 2020) and multiverse analysis (Steegen et al. 2016) formalize this by computing and reporting the *distribution* of estimates over the reasonable analyses, rather than a single preferred number. The approach has been adopted widely — in psychology, epidemiology, political science, sociology and economics — and is increasingly expected as evidence that a finding is robust to analytic flexibility, a concern sharpened by demonstrations of selective reporting in the published literature (Brodeur et al. 2016; Christensen and Miguel 2018; Simmons et al. 2011).

In practice, multiverse analyses vary three kinds of choices: which covariates to include, which observations to keep, and how to operationalize the variables. A fourth choice — which *estimator* to use — is almost always fixed by disciplinary convention. For data with a longitudinal or panel structure, where the same units are observed repeatedly, this convention is hard to defend, because the estimator is precisely the choice of *identifying assumption*. Pooled regression uses all of the

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variation in the data; a between estimator uses only differences across units in their long-run averages; a within (fixed-effects) estimator discards all cross-unit variation and identifies effects only from changes within a unit over time, thereby removing all stable, time-invariant confounding — at the cost of being unable to estimate anything that does not change within a unit; random-effects estimators occupy an intermediate position under a stronger exogeneity assumption (Mundlak 1978; Wooldridge 2010). Two estimates that differ only in this choice differ in what they are estimates *of*. If the purpose of a multiverse analysis is to characterize analytic flexibility, the estimator is therefore the first dimension that should be varied, not the one held fixed.

This paper makes that argument and quantifies it. Using a well-known wage panel, we construct the complete specification curve of three estimands across seven panel estimators crossed with covariate, sample, time-effect and inference choices, and we decompose the variance of the resulting estimate distribution by analytic dimension. The estimator dimension dominates: it accounts for the large majority of the variance of the union and marriage wage premiums, while the covariate and sample choices that a conventional robustness analysis varies account for very little; a third estimand, the return to schooling, is unidentifiable under every within estimator and so is absent from half the curve. The contribution is methodological and expository rather than a new estimator or dataset: a demonstration of *where the analytic flexibility actually lies* in longitudinal data, a principled way to read a panel specification curve through its within/between identification content, and a fully reproducible, openly implemented pipeline. We use a single canonical dataset deliberately, to isolate the role of estimator choice with maximal transparency.

Related work

The idea of reporting many analyses rather than one has two roots. In psychology and the broader sciences it appears as multiverse analysis (Steenen et al. 2016) and specification-curve analysis (Simonsohn et al. 2020); in economics it has a longer pedigree in extreme-bounds analysis, which asks whether an association survives across the set of plausible control variables (Leamer 1985; Levine and Renelt 1992; Sala-i-Martin 1997). What unites these literatures is the dimension they vary: the *set of control variables*, and more recently the operationalization, the sample, and the clustering of standard errors. The estimator itself — and with it the identifying assumption — is almost always held fixed. Our contribution is to make that dimension the object of study for panel data and to show that it dominates.

A second, complementary literature documents analytic variability by crowdsourcing: many-analyst studies give the same data and question to independent teams and record how far their results diverge, in psychology (Silberzahn et al. 2018) and in applied microeconomics (Huntington-Klein et al. 2021). These designs do capture estimator choice, but as one of many *uncontrolled* human decisions; they neither enumerate the estimator axis systematically nor isolate and quantify its specific contribution to the spread of estimates.

For the dataset used here — the Vella–Verbeek wage panel — the sensitivity of the union and marriage premiums to the pooled-versus-within contrast is long understood substantively (Card 1996; Vella and Verbeek 1998), and the data are a teaching and replication staple. We are nonetheless not aware of a prior specification-curve or multiverse analysis of this dataset, nor of one in any panel setting that treats the estimator as the primary forking path and quantifies its share of the estimate distribution. The substantive sensitivity is therefore not new; the systematic, quantified and fully reproducible framing of estimator choice as the dominant analytic fork is the contribution.

Materials and methods

Data

We use the wage panel assembled by Vella and Verbeek (1998), a balanced sample of 545 young men drawn from the U.S. National Longitudinal Survey of Youth and observed annually from 1980 to 1987 (4,360 person-years). The dataset is distributed openly in the `wooldridge` and `plm` packages for R (Shea 2024; Croissant and Millo 2008) and is a long-standing teaching and replication staple, which makes it well suited to isolating the effect of estimator choice without confounding from data-cleaning idiosyncrasies. The outcome is the natural logarithm of the hourly wage. We study three focal regressors chosen to span the identification spectrum: union membership and marital status, both of which change within individuals across years, and years of completed schooling, which does not change within any individual over this window.

Specification space and estimators

The specification space crosses five estimators with a grid of analytic choices. The five estimators are pooled ordinary least squares, the between estimator, random effects, fixed effects (the within estimator), and first differences. These are crossed with the inclusion or exclusion of four covariate blocks (a quadratic in labour-market experience, industry indicators, region indicators, and a health indicator), the inclusion or exclusion of year fixed effects, and three sample definitions (the full sample; dropping person-years reporting poor health; and trimming the outcome to its 1st–99th percentiles). This yields up to 480 identified specifications per estimand. The accompanying implementation additionally supports two estimators that we use as interactive sensitivity checks but do not enumerate in the principal grid: correlated random effects — the Mundlak device, in which person-specific means of the regressors are added to a random-effects model — and two-way (unit-and-time) fixed effects. Standard errors (which quantify the uncertainty of an estimate) are computed three ways: classical, which assumes the regression errors are independent and have constant variance; heteroskedasticity-robust (the “HC1” estimator), which relaxes the constant-variance assumption; and cluster-robust by individual (the “CR1” estimator), which additionally allows an individual’s repeated observations to be correlated over time. Because the point estimate itself is unchanged by the standard-error choice, inference is treated as a separate axis rather than as a multiplier on the curve.

Decomposing the specification curve

To quantify how the spread of estimates is apportioned among the analytic dimensions, we treat each of the identified specifications as one observation and regress the focal coefficient it produced on indicator variables for the analytic choices behind it (estimator, each covariate block, year effects and sample), then partition the total variation by an analysis of variance (ANOVA). The “sum of squares” is the standard measure of total variation — the summed squared deviations of the estimates from their mean — and each dimension’s share is the percentage of that total variation associated with the corresponding choice. Because the estimator is entered first, its reported share equals its marginal contribution, i.e. the variation in estimates explained by the estimator alone. This decomposition treats each specification as one observation and the analytic choices as the explanatory factors, so it directly answers the question “which analytic decision moves the estimate the most?”

Implementation and verification

All estimators are implemented from textbook formulas in dependency-free JavaScript so that the entire analysis runs in a web browser, and the implementation is held to a research standard of reproducibility. A self-test reproduces canonical `plm` benchmarks, and every estimate in the specification space — all 1,440 specifications across the three estimands — is re-fit independently in two languages and software ecosystems: R, using `plm` (Croissant and Millo 2008), and Python, using `linearmodels`. The pooled, between, fixed-effects and first-difference coefficients agree to better than 1×10^{-6} in R and 5×10^{-7} in Python (that is, to at least the sixth decimal place), and the random-effects coefficients to 3×10^{-4} (the third–fourth decimal place); the small residual reflects alternative implementations of the Swamy–Arora estimator, the standard procedure for estimating the relative sizes of the within-individual and between-individual error variances that random effects requires. The correlated-random-effects coefficient reproduces the fixed-effects estimate to machine precision, as the Mundlak result requires, and the joint Hausman statistic matches `plm`’s implementation. Collinear regressors (for example, a within-transformed experience term that duplicates year effects) are removed by an order-preserving rank reduction that mirrors the behaviour of standard regression software.

Results

The estimator spectrum

Figure 1 shows the focal coefficient for every identified specification, grouped by estimator, for the three estimands. The pattern is the central result. For the union and marriage premiums, the estimates decline monotonically as the estimator moves from pooled and between (which use cross-unit variation) through random effects to within and first differences (which use only within-unit variation), and the within estimators yield the only statistically non-significant estimates. For the return to schooling, the fixed-effects and first-difference columns are empty: a regressor that never changes within a unit is eliminated by the within transformation and cannot be identified by those estimators.

Variance decomposition

Table 1 reports the analysis-of-variance decomposition. For the union premium, the estimator dimension alone accounts for 84.6% of the variance of the estimate across the specification space; the four covariate blocks, year effects and sample definition together account for under 7%. For the marriage premium the estimator accounts for 56.2%, with the experience control and year effects the next largest contributors. The schooling estimand inverts the pattern in an instructive way: among the estimators that *can* identify it (pooled, between and random effects), the estimate barely moves, so the estimator share is near zero; but the within estimators cannot identify it at all, so its “fragility” is not a spread of values but a binary question of whether the design identifies the estimand.



Figure 1: The estimator “spectrum.” For each estimand, every identified specification’s focal coefficient is plotted by estimator (each point is one analysis; the horizontal bar is the median; green marks 95% confidence intervals that exclude zero). The union and marriage premiums fall monotonically from pooled/between to within/first-difference estimators — the signature of confounding by stable individual heterogeneity — while the fixed-effects and first-difference columns for the return to schooling are empty because schooling does not vary within an individual.

Table 1: Share of the variance of the focal estimate attributable to each analytic dimension (analysis-of-variance Type-I sequential sum-of-squares with estimator entered first; identified specifications, %). Rows include a final “Residual” column (within-cells noise plus interactions) and sum to 100% up to rounding. Estimator choice dominates the union and marriage premiums; for the schooling estimates that are identified at all, covariate choice dominates.

Estimand	Estimator	Experience	Industry	Region	Health	Year FE	Sample	Residual
Union pre-mium	84.6	0.0	6.1	0.0	0.0	0.1	0.5	8.7
Marriage pre-mium	56.2	6.5	1.6	0.1	0.0	6.5	1.1	28.0
Return to school-ing	0.1	90.0	0.9	0.0	0.0	1.4	3.3	4.3

Two robustness checks confirm that the ranking does not depend on the ordering convention. Type-

III (marginal, order-insensitive) SS shares for the estimator dimension are 43.1% for the union premium, 29.3% for the marriage premium, and 0.0% for schooling — lower in level, but the same dominance pattern. The LMG/Shapley decomposition, which averages a dimension’s contribution over all permutations of the analytic factors, recovers the headline numbers almost exactly (84.6% / 56.2% / 0.1%). The within-cells residual is informative in its own right: the 28% residual on the marriage row is not measurement noise but estimator-by-covariate interaction — different estimators respond differently to including, for instance, experience or year effects, which is the same mechanism we describe but operating between rather than within choice dimensions.

The implication is direct, and quantifiable. A conventional robustness analysis varies covariates and perhaps the sample while holding the estimator fixed; on these data, that analysis explores a sharply narrower slice of the analytic variation than the full multiverse. Holding the estimator at pooled OLS, varying only the covariates, year effects and sample, the union premium spans [0.118, 0.186] (an interquartile range of 0.049 across 96 specifications); the full multiverse — including the estimator axis — spans [0.017, 0.270] (interquartile range 0.090, nearly twice as wide). For the marriage premium the contrast is sharper: a controls-only interquartile range of 0.034 versus 0.141 across the full multiverse, more than fourfold. These figures are also broadly consistent with the prior literature on the union wage premium using longitudinal data: Card’s (1996) NLSY analysis reported pooled estimates near 0.15 falling to roughly 0.10 with fixed effects — values that sit comfortably inside our specification curve. The full curve (Figure 2) tells the more honest story: a union premium that is positive but ranges from roughly 0.04 to 0.26 depending on what one is willing to assume about selection, with the low, non-significant end occupied by the within estimators and the high end by pooled and between specifications.

Reading the curve through within/between content

Because the estimator dimension is the choice of identifying assumption, a panel specification curve can be interpreted rather than merely displayed. The ordering in Figures 1 and 2 is not arbitrary: estimators that retain between-unit variation sit high, those that use only within-unit variation sit low, and the gap between them is exactly the portion of the association attributable to stable, time-invariant differences between units — the classic signature of confounding by unobserved heterogeneity. A decomposition of the focal regressor’s own variation into within- and between-unit components predicts both where on the curve an estimator will land and whether an estimand is identifiable at all: a regressor with substantial within-unit variation (such as union status) supports the full spectrum, whereas one with essentially none (such as schooling here) cannot be identified by within estimators. This gives a longitudinal multiverse a structure that purely cross-sectional applications lack, and converts the question “how wide is the curve?” into the more informative “how much of the result depends on between-unit comparisons that the design cannot protect from confounding?”

Discussion

Implications for multiverse practice

Multiverse and specification-curve analyses are valuable precisely because they make analytic flexibility visible; for longitudinal data they should be directed at the dimension where that flexibility predominantly resides. Our results show that, on canonical data, estimator choice accounts for the majority of the variation in two of three estimands and determines outright whether the third can be estimated. A multiverse that varies only covariates and samples while fixing the estimator

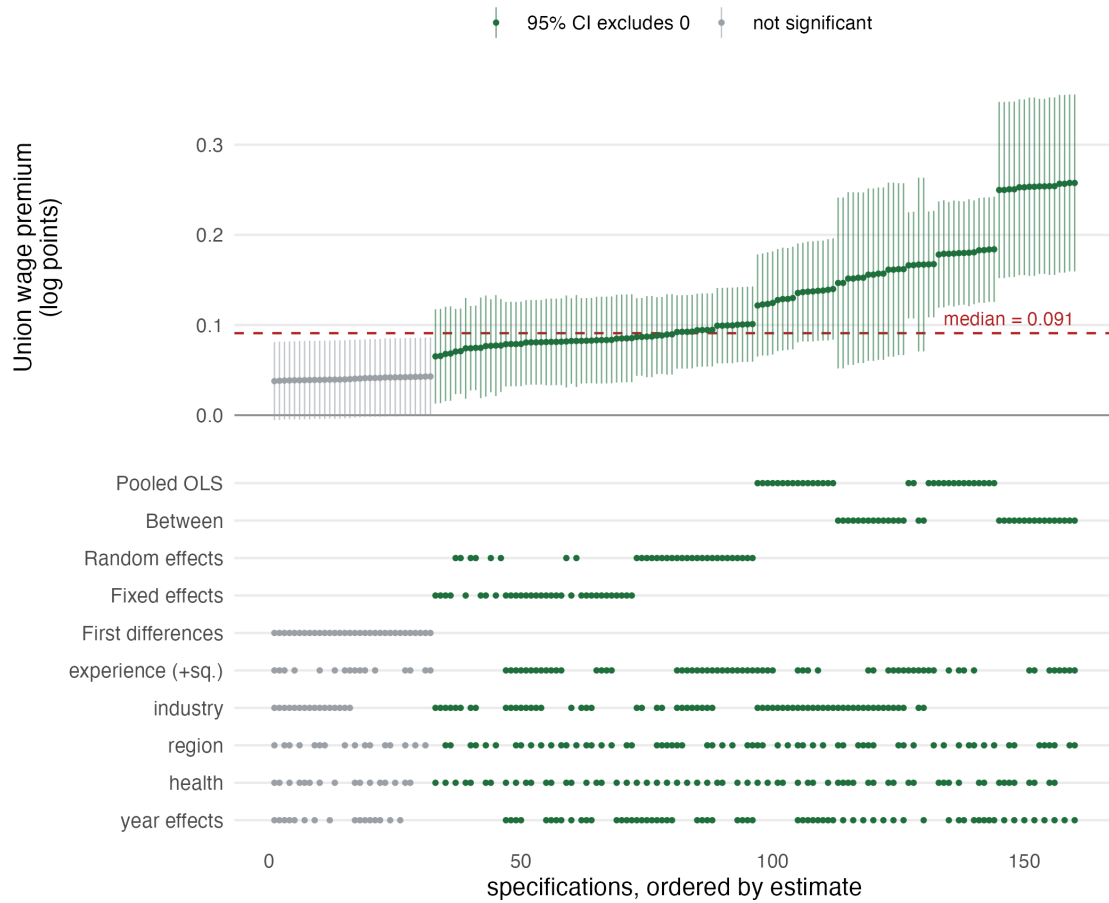


Figure 2: The full specification curve for the union premium: 160 estimates (upper panel, with 95% cluster-robust confidence intervals, sorted from smallest to largest; green marks intervals excluding zero; the dashed line is the median) above a dashboard recording the analytic choices behind each specification (lower panel). Fixed-effects and first-difference specifications occupy the low, non-significant end; pooled and between specifications the high end.

is therefore not a conservative simplification but a potentially misleading one, liable to report a narrow band of estimates that reflects an unexamined identifying assumption. We recommend that multiverse analyses of panel or longitudinal data (i) include the estimator as a first-class forking path, (ii) report which dimension drives the spread, for example via the variance decomposition used here, and (iii) interpret the curve through the within/between identification content of the estimators rather than treating the estimators as interchangeable.

A teaching application

The same apparatus serves, secondarily, as a teaching environment for identification — among the least intuitive ideas in the longitudinal-methods curriculum. The open, browser-based implementation lets students build the curves of Figures 1 and 2 themselves without installation or programming, watch a coefficient fall as the estimator moves along the within/between spectrum, observe half the curve disappear when the focal regressor is time-invariant, and inspect the equation each estimator fits. The tool also translates any specification into runnable R, Stata or Python code, supporting a transition from point-and-click exploration to reproducible scripting.

Limitations and extensions

The demonstration uses a single, deliberately simple panel; its purpose is to isolate the role of estimator choice transparently, not to establish a substantive fact about wage premiums, and the estimates should be read as illustrative. We frame the conclusion accordingly: that estimator choice dominates the multiverse spread is shown here for one canonical balanced panel and three classical estimands; whether the same dominance ranking would hold in panels with a different within/between variance ratio, in unbalanced or attrition-heavy designs, or for estimands whose covariates are more strongly correlated with each other than they are with the estimator family, is a properly empirical question we do not settle. The variance decomposition treats specifications as exchangeable observations and does not weight them by plausibility; analysts who regard some estimators as inadmissible for a given question should restrict the curve accordingly, which is itself an instance of treating the estimator as a substantive choice. Finally, the argument generalizes naturally to other designs whose credibility turns on the identifying choice — difference-in-differences, instrumental variables and event studies — each of which admits its own estimator-led multiverse and would be a productive direction for further work.

Ethics

This study used only publicly available, fully anonymized secondary data and did not involve human participants, identifiable personal data, or animals; no ethical approval was required.

Data accessibility

The analysis uses the `wagepan` dataset of Vella and Verbeek (1998), distributed openly in the `wooldridge` and `plm` R packages (Shea 2024; Croissant and Millo 2008). A tidy copy of the data, the data-preparation script, the JavaScript estimation engine, the R (`plm`) and Python (`linearmodels`) verification scripts, and the variance-decomposition script that generates table 1 are openly available at <https://kevinschoenholzer.com/sim-paneldata/> under the MIT licence, with the full source repository at [REPOSITORY URL TO CONFIRM BEFORE SUBMISSION] (the URL currently registered in CITATION.cff is <https://github.com/kevisc/panel-data-multiverse>,

which the author will confirm or replace with the public repository at submission). An archived release of the code and data extract is deposited at Zenodo with **DOI: [TO BE INSERTED FROM RESERVED ZENODO RECORD BEFORE SUBMISSION]** (10.5281/zenodo.XXXXXX), so that every figure and table can be regenerated from the archived version.

Declaration of AI use

The author used a large-language-model coding assistant to help implement the browser-based software and to edit prose. All code, analyses, numerical results, and text were reviewed and verified by the author, who takes full responsibility for the content.

Authors' contributions

K.S.: conceptualization, methodology, software, formal analysis, validation, visualization, and writing (original draft and review and editing). The author approved the final version and agrees to be accountable for all aspects of the work.

Conflict of interest declaration

The author declares no competing interests.

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The wage-panel extract derives from Vella and Verbeek (1998) and is redistributed through the `wooldridge` (Shea 2024) and `plm` (Croissant and Millo 2008) R packages.

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